

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

- Line 1 contains N1, followed by N1 integers of the first array
- Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 void solve() {
4     int t;
5     if (scanf("%d", &t) != 1) return;
6     while (t--) {
7         int n1;
8         if (scanf("%d", &n1) != 1) break;
9         int *arr1 = (int *)malloc(n1 * sizeof(int));
10        for (int i = 0; i < n1; i++) {
11            scanf("%d", &arr1[i]);
12        }
13        int n2;
14        if (scanf("%d", &n2) != 1) {
15            free(arr1);
16            break;
17        }
18        int *arr2 = (int *)malloc(n2 * sizeof(int));
19        for (int i = 0; i < n2; i++) {
20            scanf("%d", &arr2[i]);
21        }
22        int i = 0, j = 0;
23        int first = 1;
24        while (i < n1 && j < n2) {
25            if (arr1[i] == arr2[j]) {
26                if (!first) {
27                    putchar(' ');
28                }
29                printf("%d", arr1[i]);
30                first = 0;
31                i++;
32                j++;
33            } else if (arr1[i] < arr2[j]) {
34                i++;
35            } else {
36                j++;
37            }
38        }
39        putchar('\n');
40        free(arr1);
41        free(arr2);
42    }
43 }
44 int main() {
45     solve();
46     return 0;
47 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.